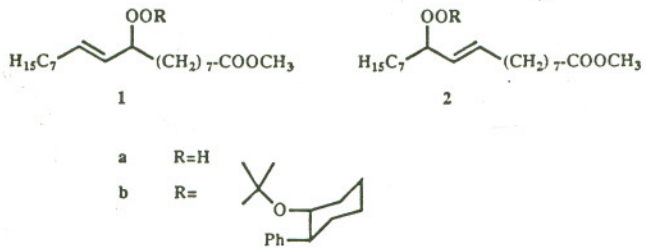
ORGANIC CUMULATIVE EXAMINATION

February 17, 1990

Recently Porter and coworkers (*J. Am. Chem. Soc.* **1990**, *112*, 1266-1267) reported a study of the rearrangements of *(R)* and *(S)* 9-hydroperoxides **1a**. *(R)* and *(S)* **1a** were prepared with optical purities >99% enantiomeric excess from their respective *(R)* and *(S)* perketal derivates **1b**. The *(R)* and *(S)* perketals were separated on normal-phase chromatography (silica with 2% ethyl acetate in hexane).

The rearrangement of **1a** to **2a** in hexane was inititated by ditert-butyl hyponitrite. The *(R)* and *(S)* 9-hydroperoxides converted, with very high stereoselectivity, to the *(S)* and *(R)* 11-hydroperoxides, respectively.



1. Assuming that both **1a** and **1b** are stable to the chromatographic medium, why did the authors choose to separate the perketals of 9-hydroperoxides **1b** on silica gel and not the 9-hydroperoxides **1a**?

2. Give a mechanism consistent for the conversion of **1b** to **2b**. Keep in mind that this mechanism should be consistent with the observed high stereoselectivity.